

① + ② will be anti-correlated



Hellings - Down curve

Real Detectors

- LVK (LIGO / Virgo / KAGRA)
 - Big interferometers
 - $L \leq 4 \text{ km}$
 - $f_{\text{aw}} \sim \mathcal{O}(1 \text{ kHz})$ "audio band"
- LISA
 - Hybrid timing / interferometer
 - $L = 2.5 \cdot 10^6 \text{ km}$
 - $f_{\text{aw}} \sim \mathcal{O}(1 \text{ mHz})$ (one cycle per half hour)
- Pulsar timing arrays
 - Timing (with millisecond pulsars)
 - $L \sim \mathcal{O}(\text{kpc})$
 - $f_{\text{aw}} \sim \mathcal{O}(1 \text{ nHz})$ (one cycle per 10 yrs)

Sources:

$$f_{\text{aw}} \sim M^{1/2} \cdot a^{-3/2}$$

$$a \propto M$$

$$\Rightarrow f_{\text{aw}} \propto 1/M$$

Big binaries
 $\Rightarrow$ low freq.

Close to merger,

$$a \sim 10M$$

(ISCO is at $6M$ for a non-rotating BH)

$$f = M^{1/2} (10M)^{-3/2} \pi^{-1}$$

$$\approx 10^{-2} \cdot M^{-1}$$

$$\approx 10^{-2} \cdot \frac{1}{M}$$

$$= 10^{-2} \left(\frac{M_{\odot}}{M} \right) \cdot 10^5 \text{ Hz}$$

$$\frac{26 M_{\odot}}{c^2/c} = 3 \text{ km} / c = 10^{-5} \text{ s}$$

$$3 \cdot 10^5 \text{ km/s}$$

$$f = 10^3 \left(\frac{M_0}{M} \right) \text{ Hz}$$

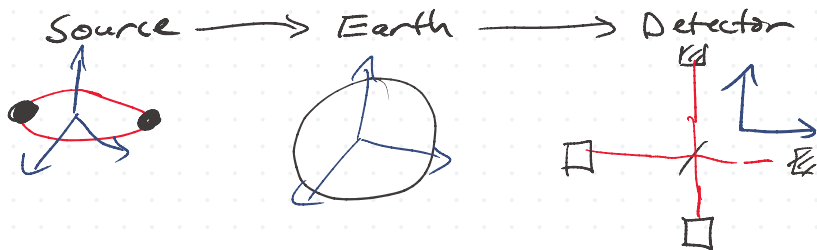
$$\left(\frac{M}{M_0} \right) \sim \left(\frac{1 \text{ kHz}}{f} \right)$$

LIGO 1 kHz $\rightarrow M \sim M_\odot$ $\rightarrow$ Dead stars
 LISA 1 mHz $\rightarrow M \sim 10^6 M_\odot$ $\rightarrow$ supermassive black holes
 PTAs 1 nHz $\rightarrow M \sim 10^{12} M_\odot$ $\rightarrow$ Don't exist.
 see inspirals instead of mergers
 $\sim 10^9 M_\odot$

GW Detector Response

LIGO Conceptually, we're changing the lengths of an interferometer $\Rightarrow$ signal!

We start in source frame, coordinate transform into the interferometer frame.



Wave frame:

$$h_{ij} = \begin{pmatrix} h_+ & h_x & 0 \\ h_x & -h_+ & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$e^+ = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \hat{1} \otimes \hat{z} - \hat{j} \otimes \hat{j}$$

$$e^x = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \hat{1} \otimes \hat{j} + \hat{j} \otimes \hat{1}$$

$$h_{ij} = h_+ e^+_{ij} + h_x e^x_{ij}$$

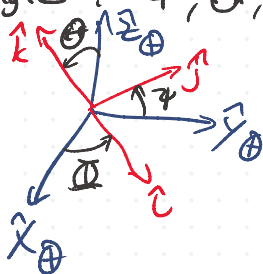
Earth frame (centered at geocenter)

$\hat{z}_{\oplus} \rightarrow \{90^\circ N, 0\}$ (points out of north pole)

$\hat{x}_{\oplus} \rightarrow \{0, 0\}$ (zero island)

$\hat{y}_{\oplus} \rightarrow \{0, 90^\circ E\}$ (Indian ocean)

Euler angles: ψ, θ, ϕ (Wikipedia for correct version)



$$R_{\psi} = \begin{pmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R_{\theta} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{pmatrix}$$

$$R_{\phi} = \begin{pmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R = R_{\psi} R_{\theta} R_{\phi}$$

Sky angles vs. Euler angles

- Source is located in Earth-frame by (lat/lon) (θ, ϕ) , observed polarization by ψ .

- Sky angles (R.A. and Dec.) (α, δ)

$$\theta = \frac{\pi}{2} - \delta, \quad \phi = \alpha - \text{Greenwich Mean Standard Time}$$

- Sky angles to Euler angles

$$\Theta = \pi - \theta, \quad \Phi = \phi - \frac{\pi}{2}, \quad \Psi = \psi$$

$$R_{\psi} = R_{\Psi}$$

$$R_{\theta} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\cos \theta & \sin \theta \\ 0 & -\sin \theta & -\cos \theta \end{pmatrix}$$

$$R_{\phi} = \begin{pmatrix} \sin \phi & -\cos \phi & 0 \\ \cos \phi & \sin \phi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Transform from
Wave frame to
Earth-fixed
coordinates

Rewrite polarization tensors

$$\hat{e} = (\sin\phi \cos\gamma - \sin\gamma \cos\gamma \cos\theta) \hat{x} \\ - (\cos\phi \cos\gamma + \sin\gamma \sin\phi \cos\theta) \hat{y} \\ + (\sin\gamma \sin\theta) \hat{z}$$

$$\hat{j} = (-\sin\phi \sin\gamma - \cos\gamma \cos\phi \cos\theta) \hat{x} \\ + (\cos\phi \sin\gamma - \cos\gamma \sin\phi \cos\theta) \hat{y} \\ + (\sin\theta \cos\gamma) \hat{z}$$

$$\hat{k} = (-\sin\theta \cos\phi) \hat{x} \\ + (-\sin\theta \sin\phi) \hat{y} \\ - \cos\theta \hat{z}$$

We want to write the strain as $h = F^+ h_+ + F^x h_x$
 F^+, F^x : antenna pattern functions

$$h = D_{ij} \dot{w}^i \quad D = \frac{1}{2} (\hat{x}_0 \otimes \hat{x}_0 - \hat{y}_0 \otimes \hat{y}_0)$$

$L \ll \lambda_{\text{gw}}$

$$F_+ = -\frac{1}{2} (1 + \cos^2\theta) \cos 2\phi \cos 2\gamma + \\ - \cos\theta \sin 2\phi \sin 2\gamma$$

$$F_x = \frac{1}{2} (1 + \cos^2\theta) \cos 2\phi \sin 2\gamma + \\ - \cos\theta \sin 2\phi \cos 2\gamma$$

Detector response for a detector aligned w/ the Earth frame.

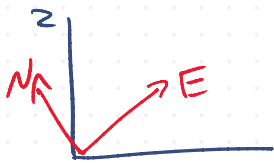
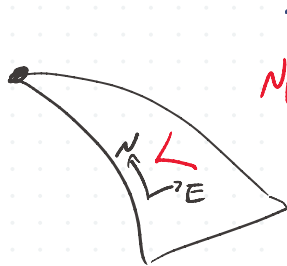
Convert to detector frame:

$$x_{\oplus} = (R+h) \cos\phi \cos\lambda$$

$$y_{\oplus} = (R+h) \cos\phi \sin\lambda$$

$$z_{\oplus} = \left(\frac{b^2 R}{a^2} + h \right) \sin\phi$$

$$R = \frac{a^2}{(a^2 \cos^2\phi + b^2 \sin^2\phi)} \frac{1}{2}$$



- γ_1 : N of E of arm 1 proj. to local horizontal
 γ_2 : Same for arm 2
 ω_1 : smallest angle between arm 1 and local hor. until
 ω_2 : same for arm 2

$\Rightarrow$ Apply rotations from detector frame to Earth frame.
 Write θ, ϕ, ψ in terms of $\phi, \lambda, h, \gamma_1, \gamma_2, \omega_1, \omega_2$

$$\Rightarrow F_I^{+X}(\alpha, \delta, \text{GMST}, \psi), \Delta t_I(\alpha, \delta, \text{GMST})$$

$\rightarrow$ 4 unknowns: $\alpha, \delta, \psi, \text{GMST}$

$\rightarrow$ 2 observables per detector: h, t

$\Rightarrow$ Need at least 2 detectors!

$\Rightarrow$ Next time localization